

CHAPTER – 01

INTRODUCTION

1.1 – Project Overview

The Electric Vehicle Charge and Range Analysis Visualization Tool is a cool interactive Tableau dashboard that helps us look at how electric vehicles are doing. It is made to help people understand things like how electric vehicles charge how well the batteries work, how energy they use and how far they can go. All of this is shown in a way that's easy to understand using pictures and graphs.

More and more people are starting to use vehicles so it is getting really important to understand what affects how well the batteries work and how well they charge. This project uses Tableau to take all the data from vehicles and turn it into things that people can actually use to make decisions. This is useful for people who own vehicles for people who manage groups of electric vehicles and for people who study them.

Objectives:

- We need to look at how people charge their vehicles and how long it takes to charge them.
- We have to see how well the batteries are working and how energy they use.
- We want to compare how different electric vehicle models can go.
- We need to figure out what things affect how well the vehicles run, like the size of the battery and what kind of charger they use.
- We will make boards that people can use to look at the data in real time.

Key Features:

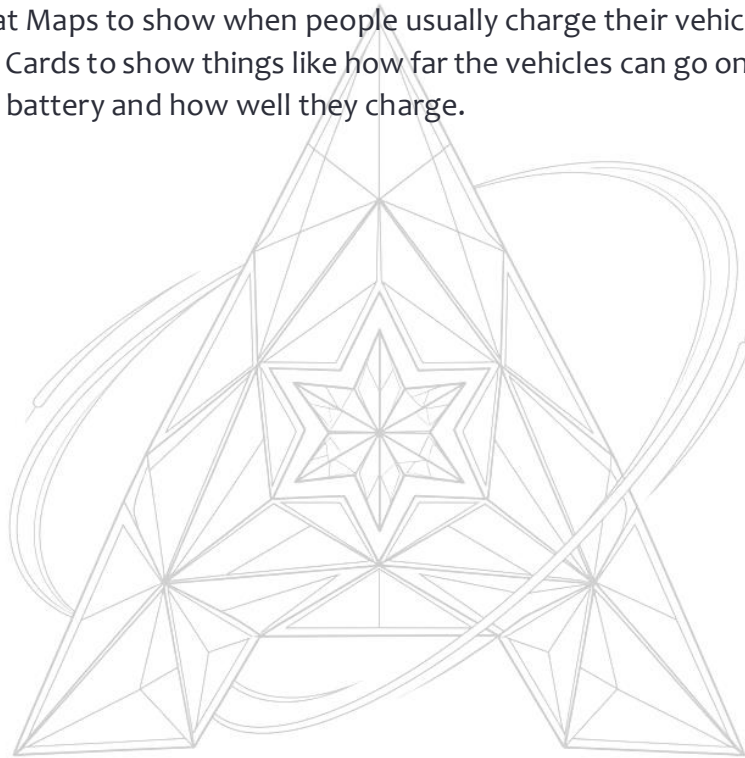
- Charge Analysis: This shows us when people charge their vehicles how they charge them and how often they charge them.
- Range Analysis: This compares how far people actually drive their vehicles to the size of the battery and how much it is charged.
- Energy Consumption Monitoring: This keeps track of how energy the vehicles use and how well they run.
- Performance Comparison: This compares electric vehicle models to see which ones go the farthest and have the best batteries.
- Interactive Filters: This lets people pick which data they want to look at, like the type of vehicle what kind of charger it uses, the date and where it's

Visualization Tool for Electric Vehicle Charge and Range Analysis using Tableau

Tableau Visualizations:

The board may have:

- Bar Charts to compare how long it takes to charge vehicles.
- Line Charts to show how energy the vehicles use over time.
- Scatter Plots to show how the size of the battery affects how far the vehicle can go.
- Heat Maps to show when people usually charge their vehicles.
- KPI Cards to show things like how far the vehicles can go on average the size of the battery and how well they charge.



ALPHA
— INNOVATORS —

